

PHY324 Blackbody Radiation Lab Report

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1 Abstract

In this experiment, we investigated the properties of a blackbody, in particular the emission of light from such a body. A blackbody is a body that absorbs light at all frequencies and emits at a specific frequency at thermal equilibrium.

We began by investigating the equipment and data processing software necessary to analyze a blackbody's radiation. Then, we investigated two properties of blackbody radiation which are Wien's Displacement Law and the Stefan-Boltzmann Law.

A close agreement was found with both laws, finding a constant of $28 \pm 1 \times 10^{-4} \text{ m} \cdot \text{K}$ for Wien's displacement law. This value is in agreement with the tabulated value of the constant, within uncertainty ranges. There was a linear relationship between the fourth power of temperature T^4 and the luminous intensity I , which is what we expect from the Stefan-Boltzmann Law. We concluded the experiment by attempting to fit the Stefan-Boltzmann model with a coefficient corresponding to the emissivity ϵ and a calibration factor which was applied to the equation.

We found a reduced χ^2 value of 2.24 and a p-value of 0.098 for this fit, as well as imperfect residuals, suggesting a poor fit. This may be explained by the large number of uncertainties in our experiment, such as an uncertain (though expected small) amount of the total luminosity reaching our spectrophotometer.

2 Introduction

Blackbody radiation is famous for having started the quantum revolution, since the Rayleigh-Jeans classical model of blackbody radiation led to the 'ultraviolet catastrophe', whereby a small enough wavelength led to an infinite intensity of light, which is physically impossible. Planck's model with a quantized unit of light solved the problem and was the first step towards quantum mechanics.

The radiated intensity of a blackbody varies according to wavelength and temperature. A higher temperature shifts the peak intensity to higher energy (higher frequency, which translates to a lower wavelength).

We began our experiment by searching for the peak wavelength after having determined our initial angle, in order to verify Equation 4.

To obtain the wavelength, we combined Snell's Law for the index of refraction within a prism (Equation 1) with the Cauchy Equation (Equation 2), the latter giving the relation between the index of refraction and the wavelength.

$$n(\theta) = \sqrt{\left(\frac{2}{\sqrt{3}}\sin\theta + \frac{1}{2}\right)^2 + \frac{3}{4}} \quad (1)$$

Where n is the index of refraction and θ is the angle of refraction.

$$n(\lambda) = \frac{A}{\lambda^2} + 1.689 \quad (2)$$

Where n is the index of refraction, λ is the wavelength and where $A=13900\text{nm}$.

Combining these two equations, we obtained our own equation for the wavelength of light emission based on the angle of refraction (Equation 3)

$$\lambda(\theta) = \sqrt{\frac{A}{\sqrt{\left(\frac{2}{\sqrt{3}}\sin\theta + \frac{1}{2}\right)^2 + \frac{3}{4}} - 1.689}} \quad (3)$$

The peak wavelength of light emission from a blackbody obeys Wien's displacement law, modelled by the following equation:

$$\lambda_{max}T = 2.898.10^3 m.K \quad (4)$$

Where λ_{max} is the peak wavelength and T is the corresponding temperature of the blackbody.

We then turned to the Stefan-Boltzmann law, which is modelled by the following equation:

$$\frac{P}{A} = I = \epsilon.\sigma.T^4 \quad (5)$$

Where P is the power, A is the area, I is the luminous intensity, ϵ is the emissivity of the body, σ is the Stefan-Boltzmann constant, and T is the temperature.

We measured the total intensity for varying voltages, thus controlling the intensity and the temperature, which both depend on the voltage.

Doing so, we measured the prefactor of the right-hand side of Equation 5, consisting of a multiple of the emissivity ϵ and of the Stefan-Boltzmann constant σ .

To obtain the temperatures for both laws, we used Equation 6:

$$T = T_0 + \frac{\frac{V/I}{R_0} - 1}{\alpha_0} \quad (6)$$

Where V is the Voltage, I is the current, and where our blackbody is a tungsten filament in a lightbulb, with, $R_0=1.1\Omega$ and $\alpha_0=4.5\times 10^{-3}.K^{-1}$ and with $T_0=293K$ at room temperature.

3 Methods and Procedures

3.1 Calibration of apparatus and initial angle

The equipment used for this experiment comes from PASCO scientific, for emission we used a blackbody light source, Model OS-8542, we used a flint glass prism, Model OS-8543 and a spectrophotometer for detection, Model OS-8534.

The setup of our experiment is illustrated in Figure 1, which is from Ref[3].

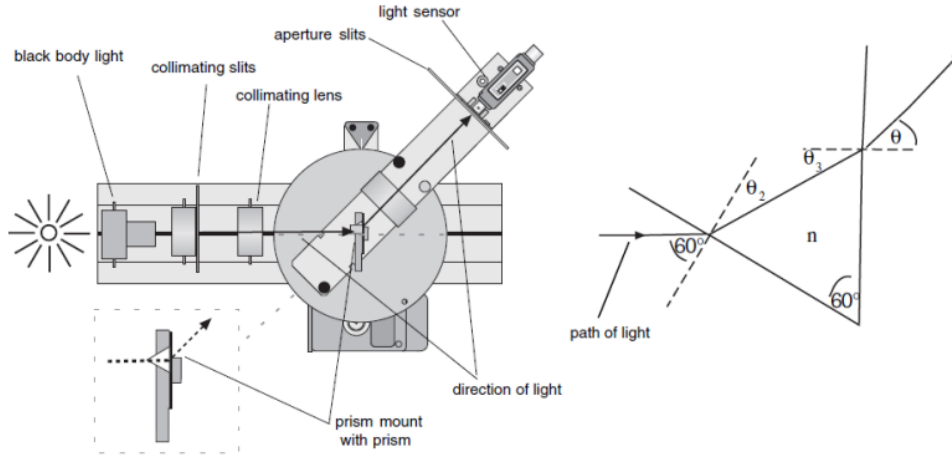


Figure 1: Setup of our Blackbody radiation experiment

The apparatus was specifically set up as follows: the collimating slits were removed from the front of the light source. We found that this provided a better spectrum of the light visually. Each part of the apparatus can be manually placed at a chosen distance - the equipment has distance labelling on it in cm. The light source was placed at 6cm, the collimating lens closest to the light source was placed at 23 cm, and the rotary motion sensor was placed at 46 cm.

Step 1: Make sure the apparatus is complete and that no unnecessary lenses are present on the optical bench. We had to remove the collimating slits from the bench, for better results.

Step 2: Turn on the light source by using the switch.

Step 3: Set the light sensor gain switch to x10 and press tare.

Step 4: Turn the rotating arm counterclockwise until it comes to a stop.

Step 5: Turn on the acquisition software, select a 90 degree acquisition and a 6V voltage and start the acquisition. Slowly rotate the arm clockwise until the acquisition comes to a stop.

Step 6: Calibrate the collimating lens and the light source positions so that the acquisition software detects one large peak and one small blip to its right.

Step 7: Note down the positions of all equipment and fix positions when possible (e.g. with a screw for the prism).

Step 8: Note down the angle at which the blip is detected, which corresponds to θ_{init} , the initial angle when the light source is aligned with the spectrophotometer.

The uncertainty is obtained by taking the maximum and minimum angle values of maximal intensity of the blip, subtracting them and dividing by two.

The central value is obtained by adding the maximum and minimum angle values and dividing by two.

Step 9: Repeat this measurement 4 times to improve accuracy and average out the central values and uncertainties to obtain θ_{init} and its uncertainty.

3.2 Wien's Displacement Law

Step 1: Switch to the 30 degree scan for more accurate measurements of the main peak and switch to a x1 gain, pressing tare, to avoid saturation of the photodetector on higher voltages.

Step 2: Select voltages of 4V, 6V, 8V and 10V and start the acquisition.

Step 3: Rotate the arm until the acquisition stops, for every acquisition.

Step 4: Measure the main peak angle θ_{peak} using cursors slightly to the left and slightly to the right of the main peak.

The uncertainty is obtained by taking the maximum and minimum angle values, subtracting them and dividing by two.

Read out the current from the acquisition software.

Take two data sets for each and all measurements for more accuracy.

Step 5: Average out the central values and uncertainties to obtain θ_{peak} and its uncertainty for each voltage.

Step 6: Determine the angular separation for each voltage by subtracting θ_{init} from θ_{peak} values.

Step 7: Convert the angular separation to a wavelength by using equation 3.

Step 8: Calculate the temperature for each voltage by using equation 6.

Step 9: Calculate the constant of Wien's displacement law by using equation 4.

Step 10: Compare the experimental constant to the tabulated lab manual constant.

3.3 Stefan-Boltzmann law

Step 1: Switch back to the 90 degree scan, for a full integrable main peak.

Step 2: Make sure to always stay at the same gain, otherwise measurements will be inconsistent.

Step 3: Make 7 sets of measurements at 4V, 5V, 6V, 7V, 8V, 9V and 10V.

Measure the area under each main peak using the cursors, by clipping off the wings of the peak and using the 'integrate under curve' button on the acquisition software.

Multiple runs would increase accuracy of results and provide better uncertainties.

Read off the current value for each run from the acquisition software.

Step 4: Calculate the temperature corresponding to each voltage using equation 6.

Step 5: Attempt to fit the emissivity ϵ exponent by curve fitting equation 5.

Since the absorption by the spectrophotometer is dependent on the solid angle and since there is significant loss in the signal, the equation will be hard to fit exactly.

Instead of searching for a perfect fit, calibrate the result for the first voltage and make sure the relationship is followed by the other data points.

Step 6: Plot temperature to the fourth power T^4 against the luminous intensity I to confirm the linear relationship. The slope of this linear relationship is ϵ times the Stefan-Boltzmann constant σ , times the calibration factor found previously.

Step 7: Compare the experimental results to the theoretical Stefan-Boltzmann law.

4 Results

Note: notes and equations used for the propagation of uncertainties in the figures below can be found in the Appendix section on the last page.

Initial angle (degrees)	Peak angle (degrees)
81 ± 2	24.5 ± 0.5
80 ± 2	23.9 ± 0.2
80 ± 2	23.38 ± 0.09
80 ± 1	23.5 ± 0.2
mean = 80 ± 4	mean = 23.8 ± 0.6

Table 1: Calculated Initial and Peak Angles - as described in 3.1 step 7 and 8 of the Methodology section

Wavelength (nm)	Temperature (K)
1450 ± 30	2200 ± 60
1040 ± 50	2590 ± 40
880 ± 40	2870 ± 40
920 ± 10	3230 ± 30

Table 2: Wavelength and Temperature Measurement Calculations - as described in subsection 3.2 of the Methodology section

Voltage (V)	Current (amps)	Total Intensity ($V \cdot \text{degrees}$)	Total Intensity (W/m^2)	Temperature (K)
4.0 ± 0.5	0.387 ± 0.001	0.9866 ± 0.0001	$4.5 \cdot 10^6 \pm 0.5 \cdot 10^6$	2150 ± 50
5.0 ± 0.5	0.438 ± 0.001	1.3276 ± 0.0001	$6.1 \cdot 10^6 \pm 0.5 \cdot 10^6$	2370 ± 50
6.0 ± 0.5	0.486 ± 0.001	2.5155 ± 0.0001	$11.0 \cdot 10^6 \pm 0.5 \cdot 10^6$	2560 ± 50
7.0 ± 0.5	0.529 ± 0.001	3.3617 ± 0.0001	$15.3 \cdot 10^6 \pm 0.5 \cdot 10^6$	2740 ± 40
8.0 ± 0.5	0.574 ± 0.001	4.1093 ± 0.0001	$18.7 \cdot 10^6 \pm 0.5 \cdot 10^6$	2880 ± 40
9.0 ± 0.5	0.613 ± 0.001	4.8441 ± 0.0001	$22.1 \cdot 10^6 \pm 0.5 \cdot 10^6$	3030 ± 30
10.0 ± 0.5	0.649 ± 0.001	5.5608 ± 0.0001	$25.4 \cdot 10^6 \pm 0.5 \cdot 10^6$	3180 ± 30

Table 3: Voltage, Current, Total Intensity, and Temperature Measurements - as described in subsection 3.3 of the Methodology section. Errors presented in the second and third column are the reading error. Errors in the fourth and fifth column are propagated errors. (See Appendix)

The initial angle was calculated to be 80.1 ± 1.8 degrees.

The constant in Weins Displacement Law was calculated to be $28 \pm 1 \times 10^{-4}$.

A calibration factor was applied to equation (5) in order to ensure that the value of epsilon is 0.45 (the theoretical value for tungsten). This is the method we used to convert our results for the integral (total intensity) from units of ($V \cdot \text{degrees}$) to units of (W/m^2). To calculate the calibration factor, a parameter was multiplied by equation (5) and epsilon was set to 0.45. Then the curvefit function in Python was used find the value of the parameter. The value of the Calibration factor is 2.1915233×10^{-6} .

Note: to convert the measured total intensity values from from step 3 in the 'Stefan-Boltzmann law' subsection in the methodology section, the integral values are divided by the calibration factor with

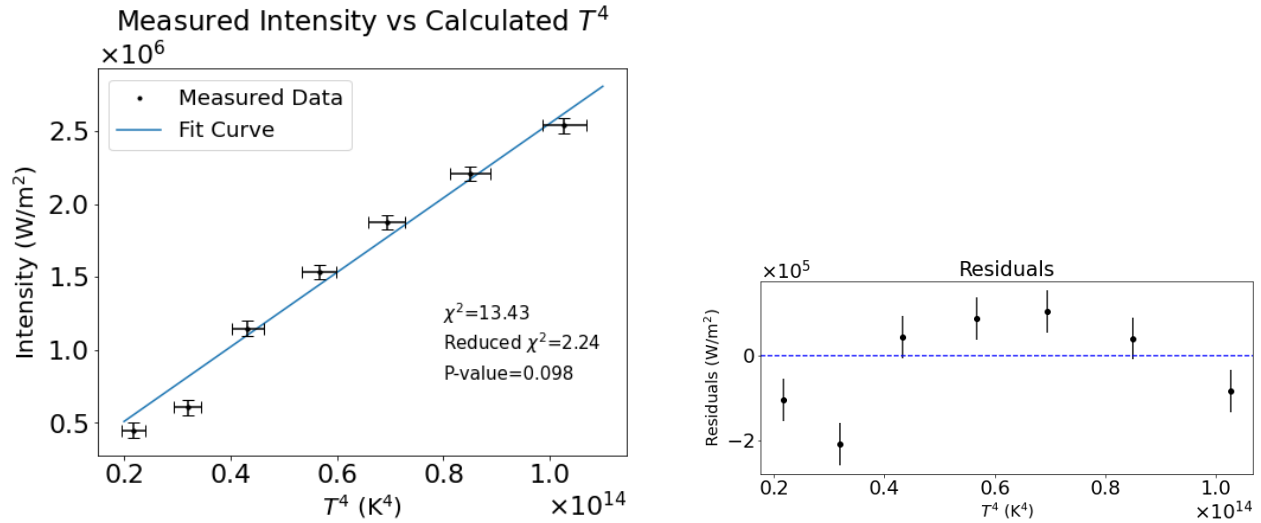


Figure 2: This is a plot of the values in the last two columns of Table 3. The uncertainties for the temperature values were propagated using equation 9 in the appendix.

units of $\frac{W}{m^2 \cdot \text{degrees} \cdot V}$. If the calibration factor is applied to equation (5) by multiplying the right hand side by 2.1915233×10^{-6} , then it is unitless.

5 Discussion

When we turn on the blackbody light source, we observe a different light depending on the voltage. A lower voltage translates to a dimmer light, more yellow and red, while a higher voltage translates to a bright white light. We also observe that all colors from red to violet are present in the spectrum of light displayed on the light sensor screen.

Based on our collected data and calculations, the constant in Weins Displacement Law was found to be 0.0028 ± 0.0001 . The value of the constant by The Committee on Data of the International Science Council (CODATA) is 0.002897771955 (where only 13 decimal places have been shown) Ref[4]. This value falls within the uncertainty range of the experimentally calculated value in this lab. This suggests that our data accurately represents the model and behaves accordingly as theoretically expected.

Data on the total intensity of light as a function of temperature was plotted in Figure 1 in order to assess whether our findings are in agreement with the Stefan-Boltzman Law. Using the curvefit function in Python, the data was fitted - with equation (5) - with a corresponding χ^2 value of 13.43 and a reduced χ^2 value of 2.24. Both of the goodness of fit values are close to 1 which suggests a reasonable fit to the model with reasonable uncertainties. However, a better fit - with χ^2 closer to 1 - can be achieved by achieving more precise estimates or reassessing (increasing) the uncertainty estimate. This is also apparent in the plot when assessing the uncertainty in the y-direction. The reduced χ^2 value is very close to one which also suggests a reasonable degree of freedom for the fit. In general, a p-value > 0.05 suggests that the model predicts the data well. The p-value is 0.098 which suggests the model reasonably represents the data. Additionally, all but two of the error bars (considering the x and y axes) in Figure 1 fall within the range of the fitted function, suggesting a good fit to the model with reasonable uncertainty values at each data point.

From the two paragraphs above, given that the collected data behaved as theoretically expected, we can also conclude that the calculated initial and peak angles in Table 1 were well assessed with reasonable uncertainties. The initial angle was found to be 80.8 ± 1.8 degrees and it is stated to be 76 degrees in the lab manual from Pasco Scientific (the manufacturer). This value does not fall within the uncertainty range but the relative error is $80.8 - 76 = 4.8$ degrees - this is a fractionally small error. Comparing it to the uncertainty in the experimentally calculated value (80.8 ± 1.8 degrees) suggests that the uncertainty of 1.8 degrees should be reassessed (it is too small).

In general, the largest sources of uncertainty are due to the propagation of reading errors. More specifically, the total intensity values in Table 3 contain the largest fractional errors. However, even these error values are reasonable and all have a fractional error of less than 5%.

6 Conclusion

We conclude that the light bulb follows blackbody radiation laws, despite being more aptly categorized as a greybody.

We found close agreement of our tungsten filament with Wien's displacement law with an experimentally calculated value for the displacement constant of 0.0028 ± 0.0001 - this falls within the range of the recommended CODATA value of 0.002897771955.

We also found a good qualitative agreement with the Stefan-Boltzmann law - the model fit crosses 5 of the 7 data points. Quantitatively, the reduced χ^2 value of 2.24 and p-value of 0.098 suggest that the uncertainties should be reassessed (increased) or the precision of the methods should be increased. The trend in our data points explains why the color of our light bulb shifts towards a whiter color at a higher voltage, since the temperature is also higher and the peak therefore shifts towards more blue, whereas at lower voltage and lower temperature, the peak is towards the red and yellow, which is why we observe a red-yellow light. This is also why at the highest temperature, there is more luminous intensity in the visible part of the light spectrum rather than in the infrared region of the spectrum.

Following these characteristics, we infer that a light bulb could be made more efficient by being at a hotter temperature so that it emits more light in the visible part of the light spectrum. However, this is to be taken with a grain of salt since a higher temperature would also mean more heat loss by the Joule effect, which would mean less efficiency. The search for optimal efficiency for a filament light bulb is therefore conditioned by that trade-off. Lastly, it might also be the case that the temperature of maximum efficiency is one that melts the filament.

Appendices

A Uncertainty Calculations

For additions and subtraction,

$$\Delta z = z \cdot \sqrt{(\Delta x)^2 + (\Delta y)^2} \quad (7)$$

Equation (7) was mainly used to propagate uncertainties when calculating the angular separation ($\theta_{initial}$) - (θ_{peak}).

For multiplication and division,

$$\Delta z = z \cdot \sqrt{\left(\frac{\Delta x}{x}\right)^2 + \left(\frac{\Delta y}{y}\right)^2} \quad (8)$$

Equation (8) was mainly used to propagate uncertainties in the temperature calculations from equation (6) and the uncertainty in the constant value calculation in equation (4).

For powers of n,

$$\Delta z = z \cdot n \cdot \frac{\Delta x}{x} \quad (9)$$

Equation (8) was mainly used to propagate uncertainties for the T^4 values in Figure 1.

The general formula for a function f(x),

$$\Delta z^2 = \left(\frac{\partial f(x)}{\partial x} \Delta x \right)^2 \quad (10)$$

Equation (9) was used to propagate the uncertainties of the wavelength values in Table 2 which were calculated using equation (3).

References

- [1] PHY324 Blackbody Lab Manual (2023)
University of Toronto - All equations are from lab manual.
- [2] PASCO Instruction Manual and Experiment Guide for the Blackbody Radiation
https://www.pasco.com/file_downloads/Downloads_Manuals/Black-Body-Light-Source-Basic\protect\discretionary{\char\hyphenchar\font}{\{}}Optics-Manual-0S-8542.pdf
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